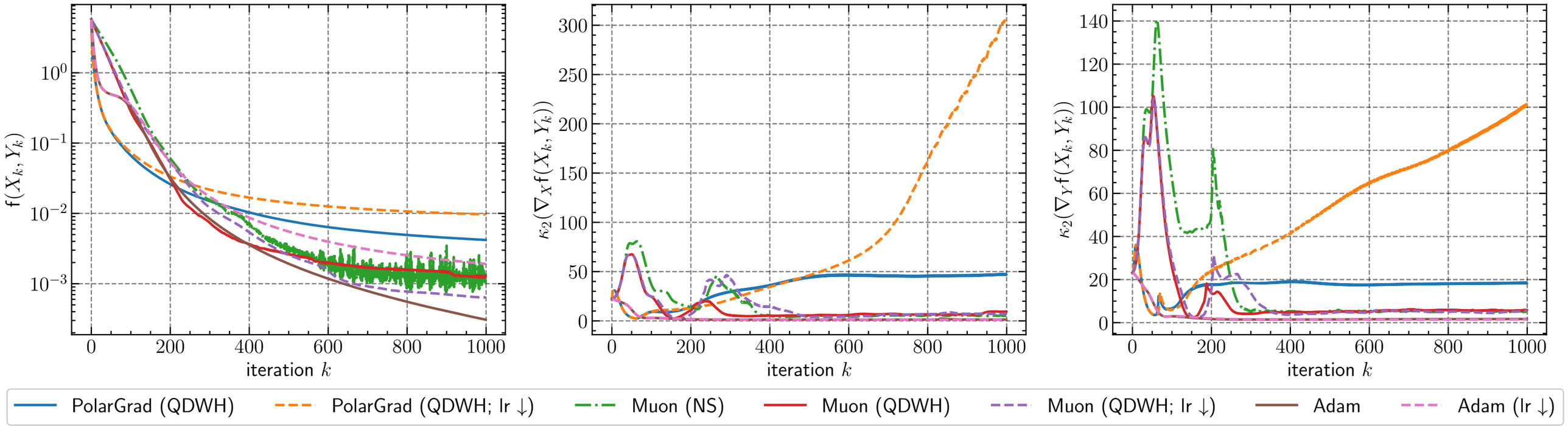
10^0
10^{-1}
10^{-2}
10^{-3}
$\mathsf{f}(X_k, Y_k)$
0
200
400
600
800
1000
iteration k
300
250
200
150
100
50
0
$\kappa_2(\nabla_X \mathsf{f}(X_k, Y_k))$
iteration k
140
120
100
80
60
40
20
0
$\kappa_2(\nabla_Y \mathsf{f}(X_k, Y_k))$
iteration k
PolarGrad (QDWH)
PolarGrad (QDWH; lr ↓)
Muon (NS)
Muon (QDWH)
Muon (QDWH; lr ↓)
Adam
Adam (lr ↓)